Fundamentals of AI and Data Science

Cybersecurity Network Traffic Analysis

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Executive Summary

This project analyses a cybersecurity network traffic dataset to investigate patterns that can be used to tell apart malicious activity from normal network traffic and predict how long the network connections will last. The analysis was conducted from the perspective of a financial organisation that has experienced a significant amount of financial losses as a result of cyberattacks. The objective was therefore not only to develop predictive models, but also to understand the network data and interpret the results within a cybersecurity context.

The dataset contained 25,192 network traffic records and 44 original features. These included connection duration, protocol type, network service, source and destination bytes, login activity and network error rates. Initial exploration found no missing values. There were 13,449 normal connections and 11,743 attack connections, meaning there was a fairly even amount of normal and attack traffic. However, the individual attack categories were not as evenly distributed.

Before training the models, categorical data such as protocol type, service and flag was converted into numerical values. Two additional features, `total_bytes` and `byte_difference`, were also created. Decision Tree and Random Forest algorithms were compared for classification and regression.

For attack classification, the Decision Tree achieved 99.58% accuracy on the test data, while Random Forest achieved 99.80%. Five-fold cross-validation produced mean accuracies of 99.61% and 99.75%, respectively. For predicting connection duration, Random Forest also performed better, achieving an R^2 score of 0.876 compared with 0.845 for Decision Tree regression.

Overall, both models performed very well when using this dataset, with the Random Forest giving the best results for both classification and regression. However, the results should still be considered alongside the limitations of the dataset, particularly the small number of examples for some attack categories.

1. Introduction and Business Understanding

Cyberattacks represent a significant risk to financial organisations because successful attacks can result in financial loss, compromised customer information, service disruption and reputational damage. Analysing network traffic can assist security teams in identifying suspicious behaviour before it causes significant harm.

Machine-learning techniques can support this process by identifying patterns across large amounts of network traffic. In the scenario given for this project, the financial organisation has previously experienced cyberattacks resulting in millions in losses. The main aim of the analysis was therefore to investigate whether information from the network traffic could be used to tell the difference between potentially malicious and normal activity.

Two prediction targets were investigated. The first was `is_attack`, a binary classification variable identifying traffic as normal (`0`) or malicious (`1`). The second was `duration`, which was treated as a numerical prediction target and therefore required a regression approach.

The project followed the CRISP-DM framework, which included understanding the business problem, exploring and preparing the data, training the models and evaluating the results **(Verma, 2026a)**. The aim was not just to find the model with the highest score, but to also understand the dataset, compare the different models and consider what the results could mean in a cybersecurity setting.

2. Data Understanding and Exploration

The dataset was imported into a Google Colab notebook using the Python pandas library. Initial examination showed that the dataset contained 25,192 observations and 44 original columns. The first five records were viewed to get an initial look at the data.

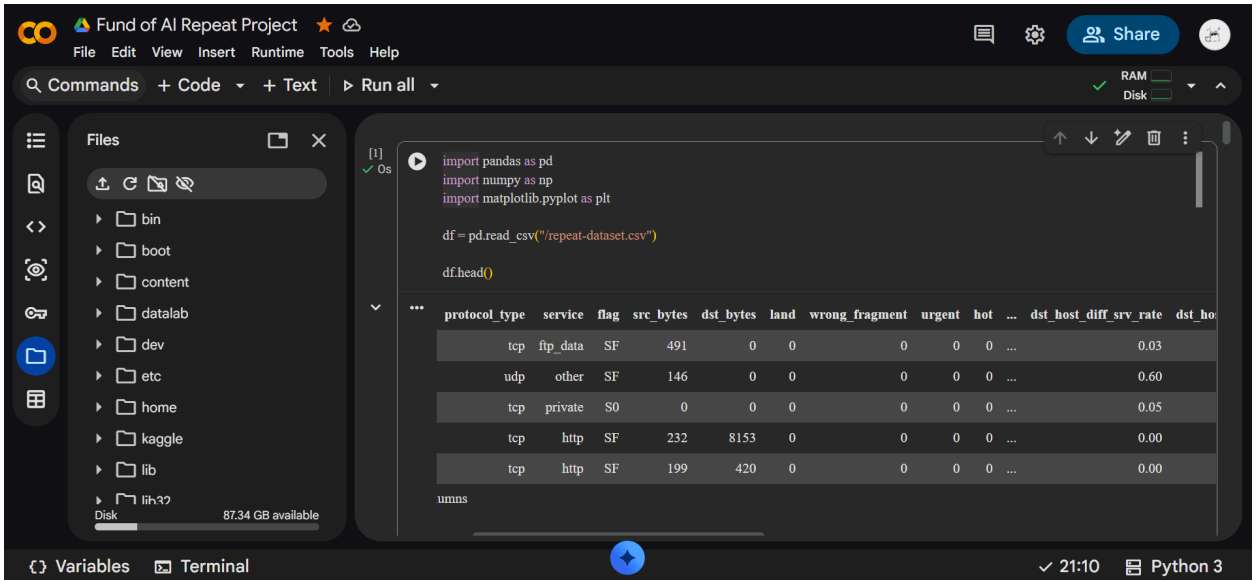


Figure 1. Sample records from the network traffic dataset.

The dataset contained both categorical and numerical data. Categorical variables included `protocol_type`, `service` and `flag`, while numerical variables included `duration`, `src_bytes`, `dst_bytes`, connection counts and several network error rates. The dataset also contained `label`, `is_attack` and `attack_category`, which provided information about whether traffic was normal or malicious.

The data was also checked for missing values before modelling. The missing value check returned zero for every column, meaning there were no missing values that needed to be dealt with.

The distribution of the `is_attack` target was then examined. There were **13,449 normal network connections and 11,743 attack connections**, which is approximately 53.4% normal traffic and 46.6% attack traffic.

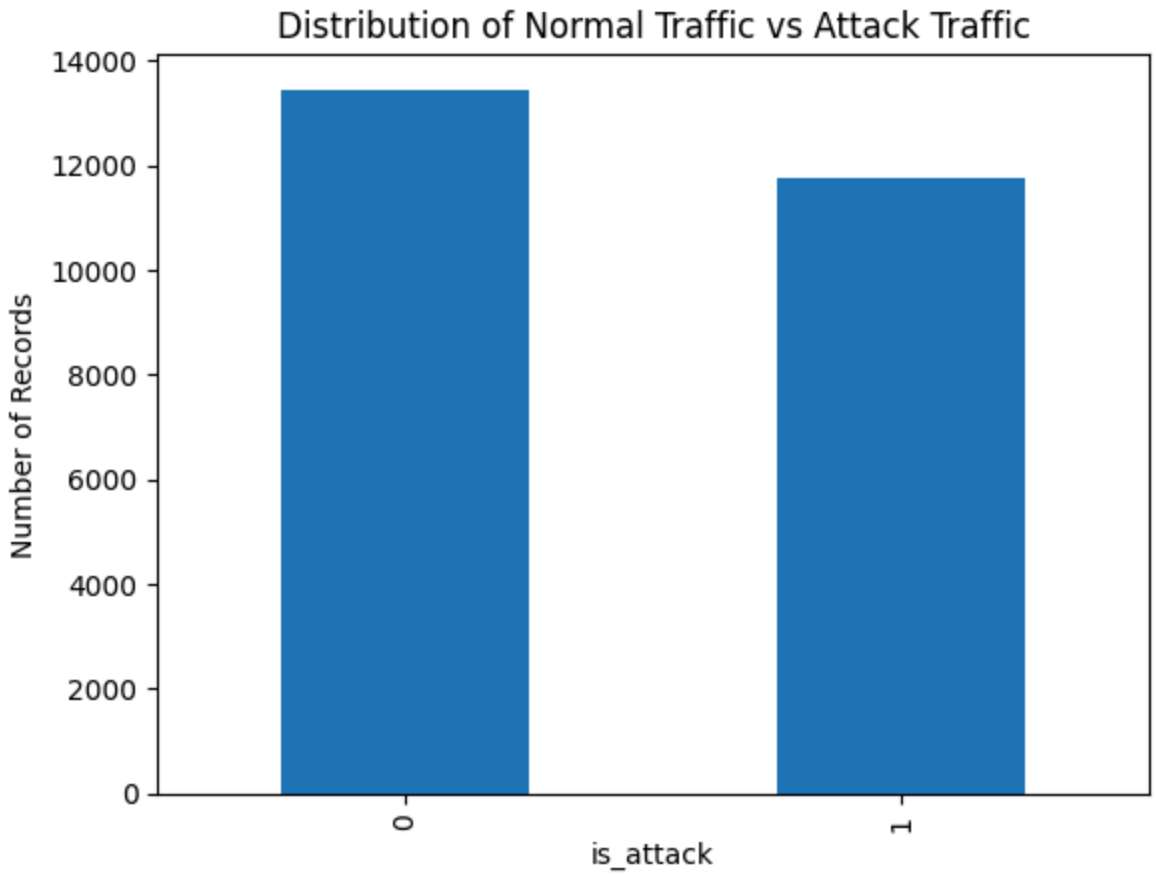


Figure 2. Distribution of normal and attack network traffic.

The number of normal and attack connections was fairly similar, with 13,449 normal connections and 11,743 attack connections. This means that neither category heavily outweighs the other when the models are being evaluated. However, when the individual types of attacks were examined, there was a much larger difference between the number of examples in each category.

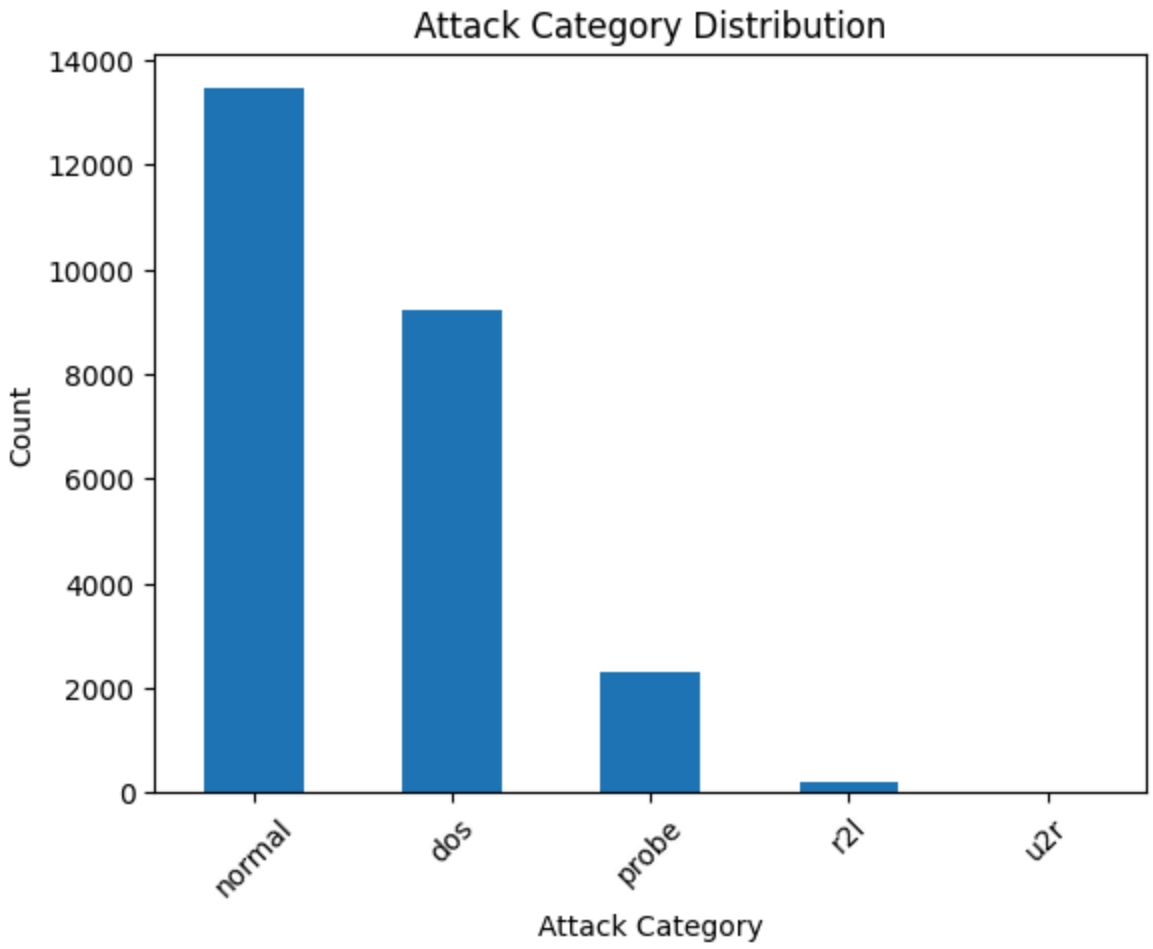


Figure 3. Distribution of network traffic by attack category.

The dataset contained 9,234 DoS attacks, 2,289 Probe attacks, 209 R2L attacks and only 11 U2R attacks. This shows that some attack types appeared much more often in the dataset than others. Although these different attacks are grouped together as attacks when predicting `is_attack`, the models had far more examples of some attack types to learn from than others.

In particular, there were only 11 U2R attacks, meaning that this type of attack was not well represented within the dataset. Because of this, in a real cybersecurity environment a model performing well overall does not necessarily mean that it would perform equally well on every type of attack, especially those with very few examples.

Relationships between the numerical variables were also explored using a correlation heatmap.

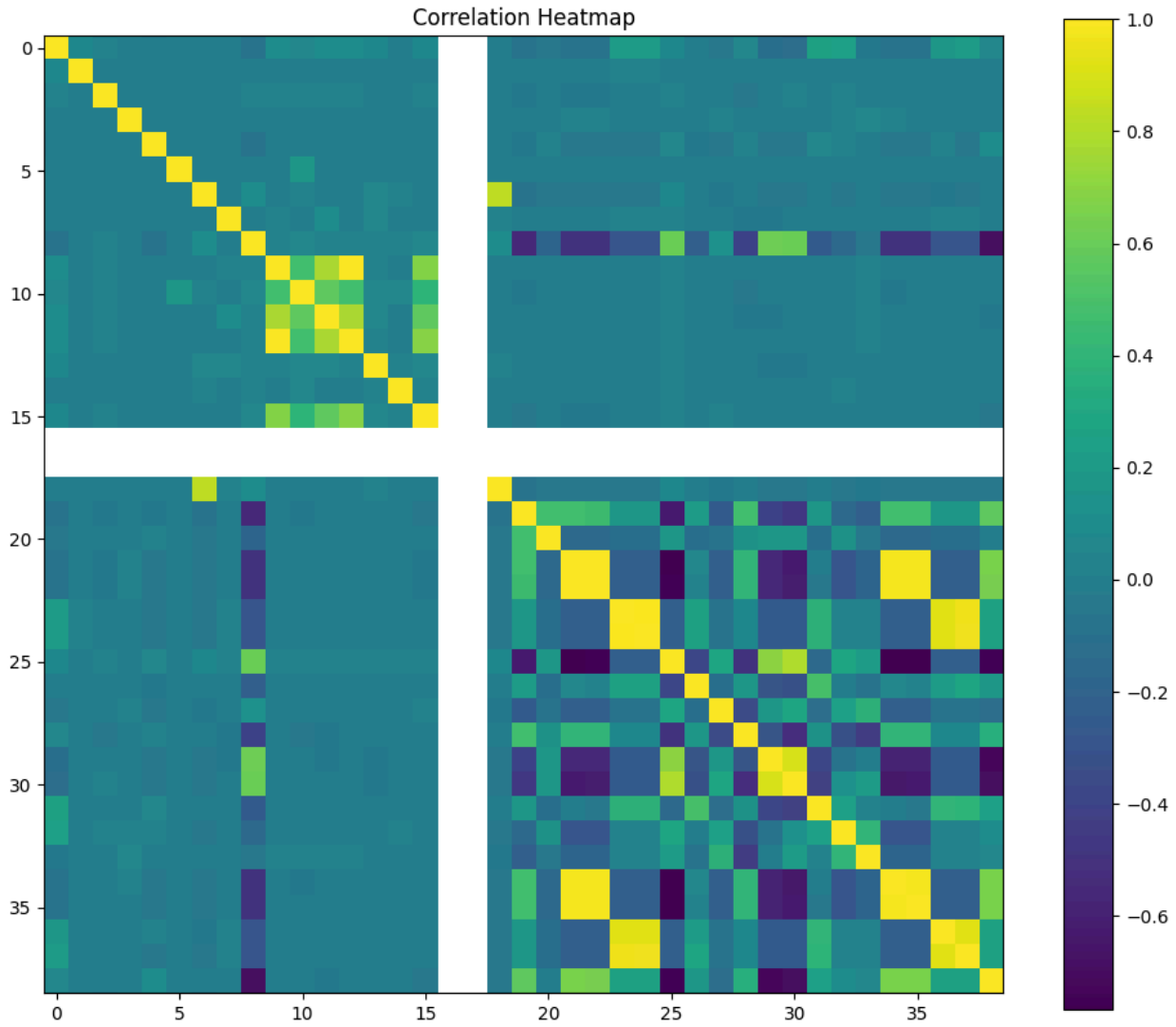


Figure 4. Correlation heatmap of numerical network traffic features.

The correlation heatmap was used to show how strongly the numerical variables were related to each other. This helped to identify which variables had stronger relationships and gave a better understanding of the dataset.

3. Data Preparation and Feature Engineering

Although no missing values were identified, the data still needed some preparation before it could be used to train the machine-learning model.

The categorical variables `protocol_type`, `service` and `flag` contained text-based categories, so these were converted into numerical values that could be used by the model **(Verma, 2026a)**.

Two additional features were created from the existing columns as part of feature engineering. The first, `total_bytes`, combined `src_bytes` and `dst_bytes` to show the total amount of data transferred during a connection. The second, `byte_difference`, showed the difference between the source and destination bytes.

These features were created to provide some additional information about the amount of data being transferred during each connection. `total_bytes` shows the overall amount being transferred, while `byte_difference` shows how different the source and destination byte values are.

For the classification task, `is_attack` was selected as the target variable. The prepared data was divided using an 80/20 train-test split. This produced 20,153 training observations and 5,039 testing observations.

The training data was used to train the models, while the testing data was kept separate and used to evaluate them afterwards. This allowed the models to be tested on data they had not already been trained on.

Five-fold cross-validation was also used to evaluate the classification models. This tested the models across five different splits of the data instead of relying only on the original train-test split.

4. Modelling and Evaluation Methodology

Decision Tree and Random Forest algorithms were compared during the modelling stage. Classification and regression versions of these algorithms were used because the project involved predicting both a category and a numerical value.

For classification, Decision Tree and Random Forest classifiers were trained to predict the `is_attack` target. A Decision Tree makes a prediction through a series of decisions based on the features in the dataset **(Verma, 2026b)**. Random Forest uses multiple decision trees and combines their predictions to produce a final result **(Scikit-learn Developers, n.d.-a)**.

Classification performance was evaluated using accuracy, precision, recall and F1-score, along with confusion matrices. Accuracy shows how many predictions were correct overall. Precision shows how often a positive prediction was correct, while recall shows how many of the actual

positive cases the model correctly identified. F1-score combines precision and recall into a single score **(Verma, 2026a)** .

Recall is particularly relevant within cybersecurity because a false negative represents malicious network activity incorrectly classified as normal. Such an error could potentially allow an attack to remain undetected.

Five-fold cross-validation was also used for both classifiers to check whether they continued to perform well across different splits of the dataset **(Scikit-learn Developers, n.d.-b)**.

For regression, Decision Tree Regressor and Random Forest Regressor models were used to predict `duration` **(Verma, 2026c)**. Their performance was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE) and R^2. MAE shows the average size of the prediction errors, while MSE gives more weight to larger errors. R^2 shows how well the model explains the differences in connection duration **(Scikit-learn Developers, n.d.-c)**.

5. Results and Interpretation

5.1 Classification Results

The Decision Tree classifier achieved an accuracy of approximately 99.58% on the 5,039 records in the testing data. The precision, recall and F1-scores were also very high for both normal and malicious traffic.

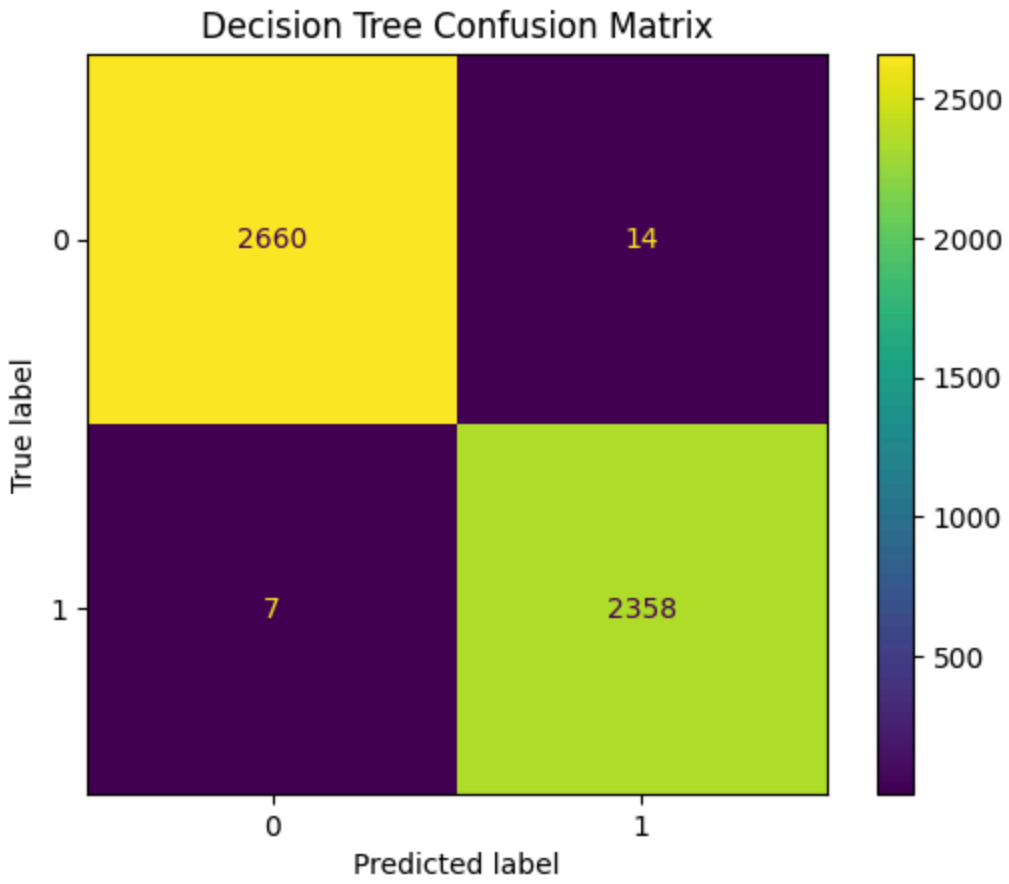


Figure 5. Decision Tree confusion matrix for attack classification.

The Decision Tree confusion matrix contained 2,660 true negatives (normal connections) and 2,358 true positives (attack connections). It incorrectly classified 14 normal connections as attacks and 7 attack connections as normal.

The seven attacks that were incorrectly classified as normal are false negatives. This is important in cybersecurity because a false negative could mean that malicious activity is not detected.

False positives can also pose a problem because normal network activity may be incorrectly flagged as an attack. However, a false negative can create a greater security risk because an actual attack has been missed.

The Random Forest classifier performed slightly better overall, achieving approximately 99.80% on the testing data.

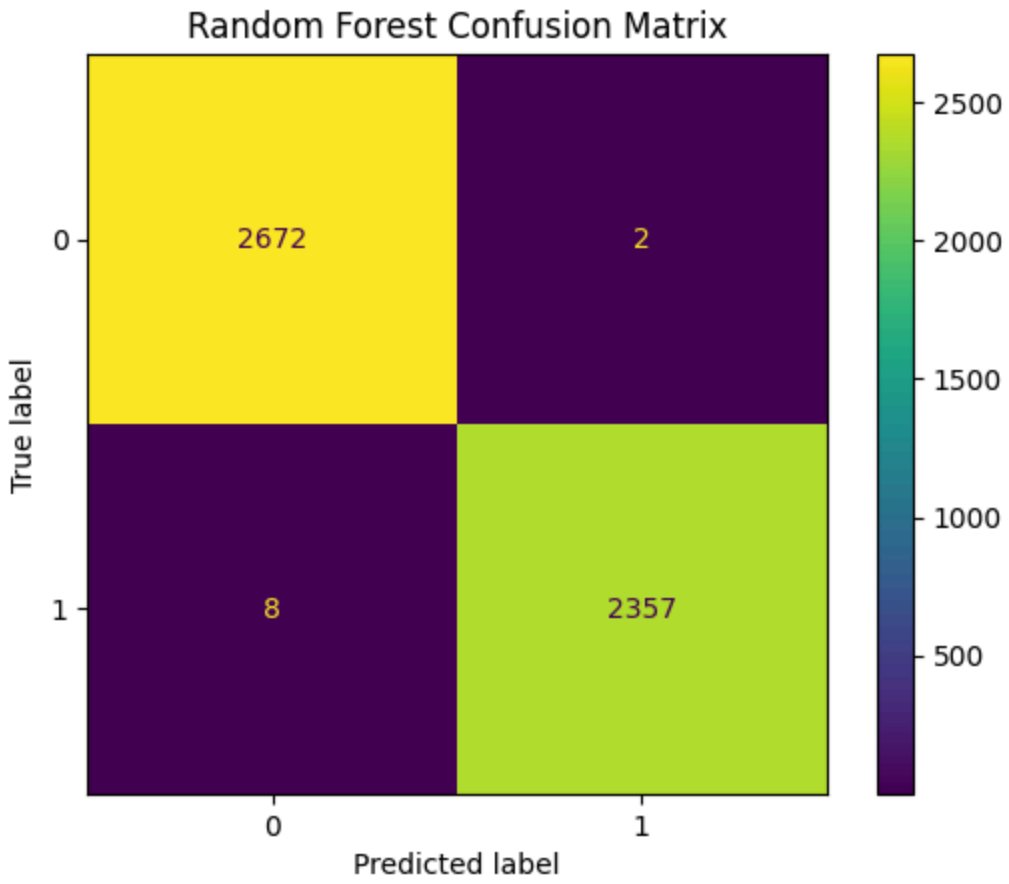


Figure 6. Random Forest confusion matrix for attack classification.

Both algorithms performed very well on the testing data, although Random Forest achieved the higher accuracy.

Five-fold cross-validation was then used to further compare their performance.

Table 1. Classification model performance comparison.

Model	Hold-out Accuracy	Cross-validation Accuracy
Decision Tree	0.995833	0.996110
Random Forest	0.998015	0.997459

The Decision Tree achieved an average cross-validation accuracy of approximately 99.61%, while Random Forest achieved approximately 99.75%. These results are very similar to the

accuracy scores from the original testing data, showing that both models continued to perform well when the different splits of data were used.

Random Forest achieved the best classification results, although the difference between the two models is small.

These high accuracy results should still be considered carefully. The `is_attack` target groups all attack types into one attack class, while the earlier analysis showed that some individual attack categories had far fewer examples than others. Therefore, a high accuracy for normal vs attack classification does not mean that the model would perform equally well for every type of attack.

This represents an important consideration for a financial organisation. Real network traffic could include rare attacks, new types of attacks or patterns that were not included in the training data. The model would therefore need further testing on different network traffic before it could be relied on in a real environment.

5.2 Regression Results

The second modelling task involved predicting network connection `duration`. Decision Tree Regressor and Random Forest Regressor were compared using MAE, MSE and R^2.

Table 2. Regression model performance comparison.

Model	MAE	MSE	R^2
Decision Tree Regressor	101.97	836,494.34	0.8454
Random Forest Regressor	99.44	670,560.92	0.8760

The Decision Tree Regressor achieved an R^2 score of approximately 0.845, while Random Forest achieved approximately 0.876. The higher R^2 score means that Random Forest performed better at explaining the differences in the connection duration in the testing data.

Random Forest also had a lower MAE of approximately 99.44 compared with 101.97 for Decision Tree. Its MSE was also lower at approximately 670,561 compared with 836,494 for Decision Tree.

Random Forest therefore performed better across all three regression measurements.

The regression results were not as high as the classification results. This suggests that predicting the exact duration of a network connection was more difficult for the models than predicting whether network traffic was normal or an attack.

Within a cybersecurity context, connection duration could still provide useful information when analysing network behaviour. However, the predictions would not always be exact, so the regression model should not be used on its own to decide whether network activity is suspicious.

6. Conclusion and Reflection on Learning

This project applied AI and data-science techniques to a cybersecurity network traffic dataset to identify malicious traffic and predict network connection duration.

The dataset contained 25,192 observations and 44 original features with no missing values. Although the number of normal and attack connections were fairly balanced, there were large differences between the individual attack categories. DoS attacks were very common in the dataset, while categories such as U2R had very few examples.

Before training the models, categorical data was converted into numerical values and two additional features, `total_bytes` and `byte_difference`, were created. Decision Tree and Random Forest were then compared for both classification and regression.

Random Forest achieved the best results overall. For classification, it achieved approximately 99.80% accuracy on the testing data and 99.75% average cross-validation accuracy. For regression, Random Forest achieved an R^2 score of approximately 0.876 compared with approximately 0.845 for Decision Tree.

Despite the model's strong results, the project also showed why the dataset itself needs to be considered when evaluating a model. Some attack categories contained very few examples, and good performance on this dataset does not guarantee that the model would perform equally well on new or previously unseen attacks.

One of the main things I learned from this project was that model performance should not be judged using accuracy alone. The dataset quality, distribution of the data, other evaluation measurements and the consequences of different types of prediction errors that are made by the model should also be considered. For example, a false negative could result in an actual attack being incorrectly classified as normal.

Using both the original train-test split and cross-validation also helped to compare the models in more than one way. Overall, this project helped me understand how data can be prepared, explored and used to train machine-learning models.

References

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Python Notebook

Public Google Colab notebook: Fund of AI Repeat Project / https://colab.research.google.com/drive/1gtVBV9WJ_RXSz46nlvDnRNY9m3N8q_5u?usp=sharing